



化学测量学—人工智能化学分析

Chapter X

Representation of Molecules



人工智能化学分析

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Contents

- 1. Quantitative Structure-Activity Relationship**
- 2. Organic Physical Chemistry**
- 3. Graph-based model and Language model**
- 4. Pre-trained molecular models**



1. Quantitative Structure-Activity Relationship

OECD Principles for machine learning

- **A defined endpoint**
- **An unambiguous algorithm**
- **A defined domain of applicability**
- **Appropriate measures of goodness-of-fit, robustness and predictivity**
- **A mechanistic interpretation, if possible**

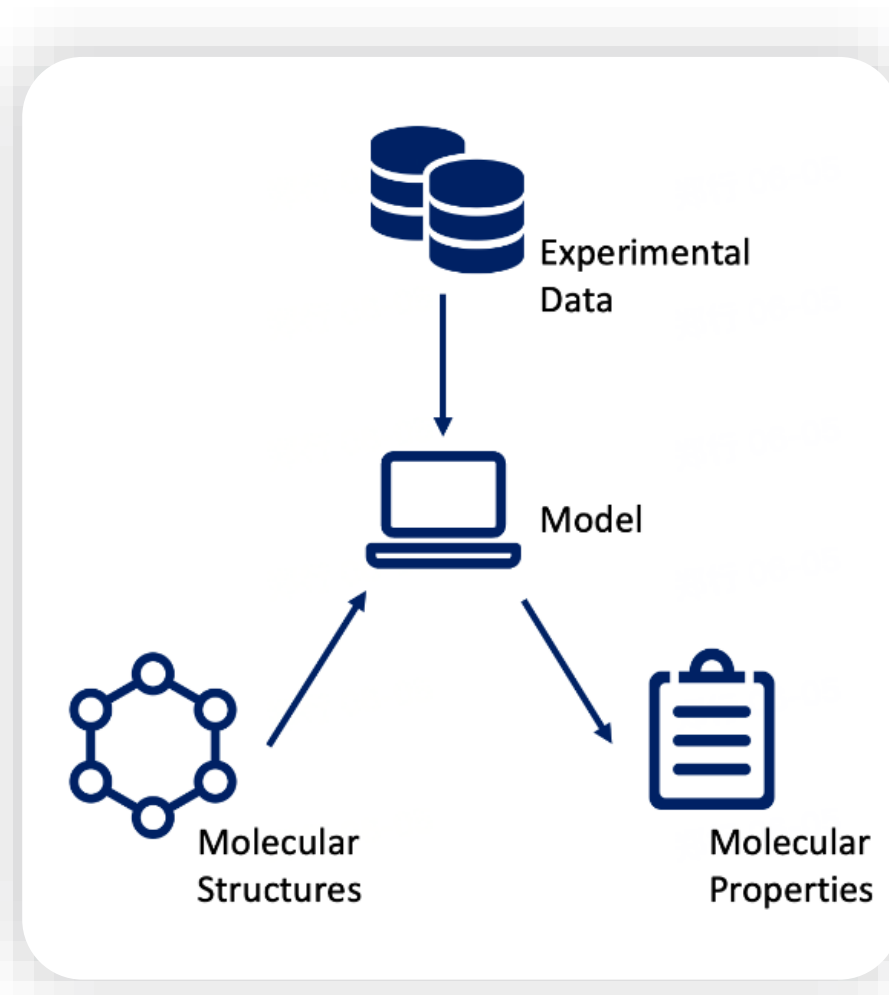
QSAR

Quantitative Structure-Activity Relationship
(QSAR) :

a method for studying the quantitative relationship between the **chemical structure** of compounds and their biological activity or **physical property** (QSPR)

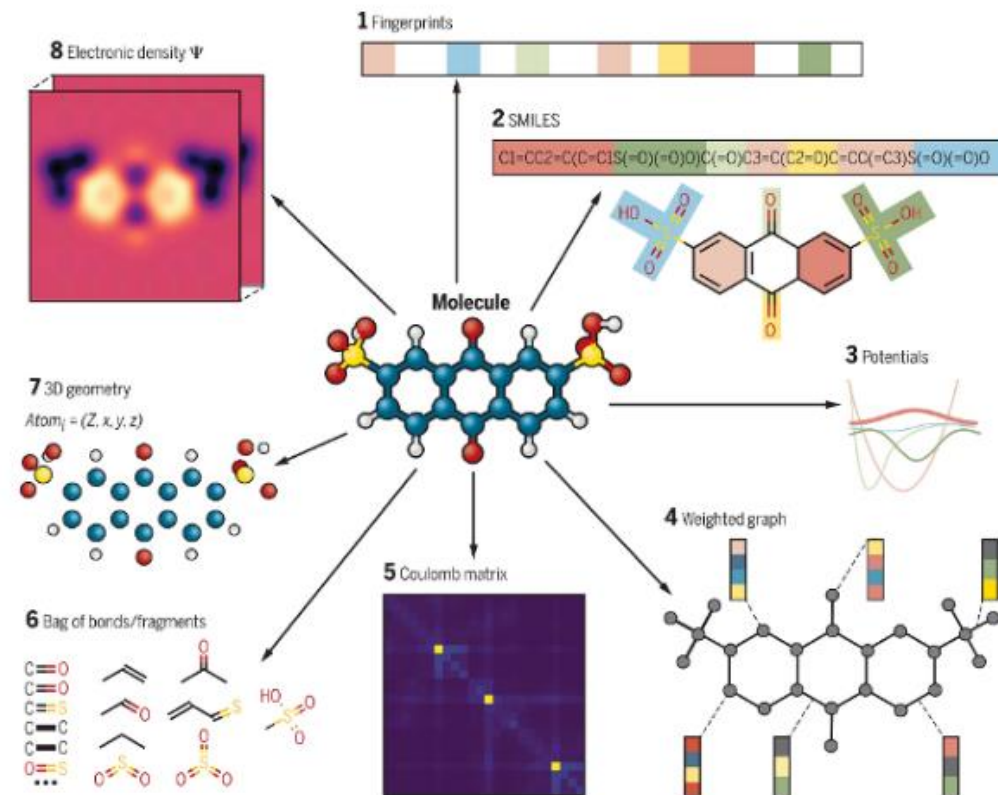
Basic Steps in QSAR

- Develop a reasonable molecular representation that transforms molecular structure into a numerically readable format by computers;
- Select a suitable machine learning model and train the model using existing molecular-property data;
- Use the trained machine learning model to predict the properties of molecules with undetermined properties.



Representation of Molecules

- 1D representation
- 2D representation
- 3D representation
- 4D representation

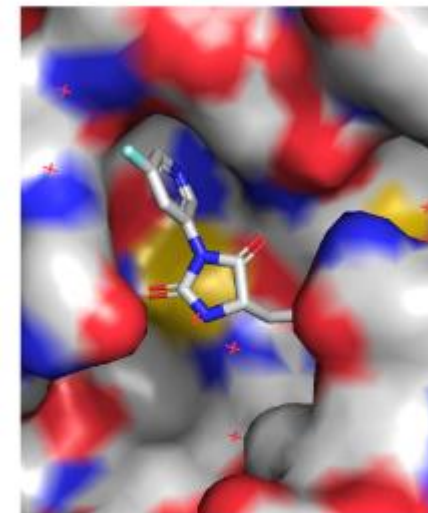


Wei, J., Chu, X., Sun, X. Y., Xu, K., Deng, H. X., Chen, J., ... & Lei, M. (2019). Machine learning in materials science. *InfoMat*, 1(3), 338-358.

Type	Description	Representation	Models
1D-QSAR	Affinity is correlated with global molecular properties of ligands, that is one value per property and ligand (pKa, logP, etc.)	• Molecular Properties	• LR • SVM • RF, DT, GBDT
2D-QSAR	Affinity is correlated with structural patterns (connectivity fingerprints, 2D pharmacophore, etc.) without consideration of an explicit 3D representation of these properties.	• SMILES • Fingerprints • Molecular Graph • Pharmacophore	• LR, SVM, RF, DT, GBDT • MLP, RNN, CNN, GNN
3D-QSAR	Affinity is correlated with 3D structure of the ligands.	• Force Field • Electronic Density • Molecular Graph with Geometry	• CoMFA • CNN • GNN

Medicinal chemistry origin of QSAR

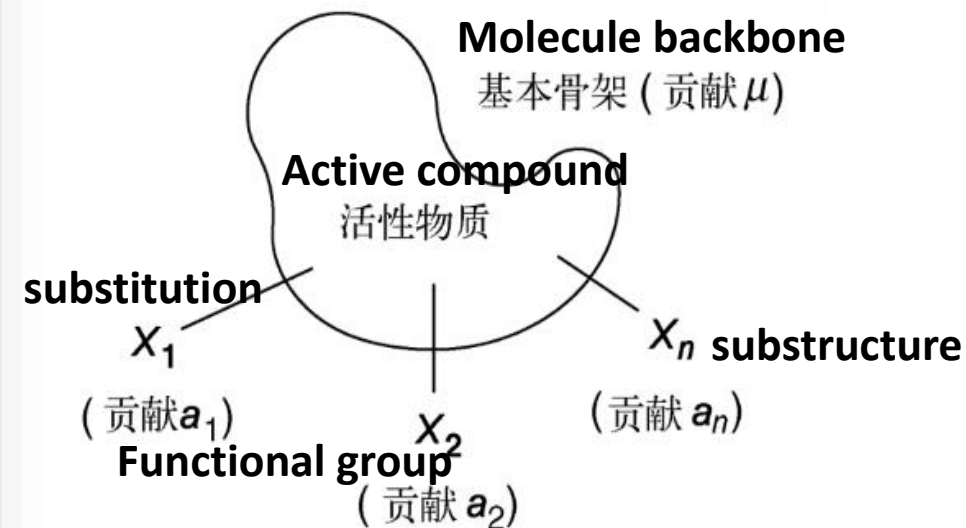
- Paul Ehrlich (1854-1915), a German chemist, proposed the “lock-key” hypothesis, which states that the interaction between a compound (key) and a biological target (lock) depends on their spatial compatibility, giving birth to a new research field of finding structure-affinity relationship (SAR).
- To go beyond experience and intuitive judgment in the beginning of SAR, scientists began to use mathematical and statistical methods to quantitatively analyze the relationship between **molecular structure** and **biological activity** in the 1960s.



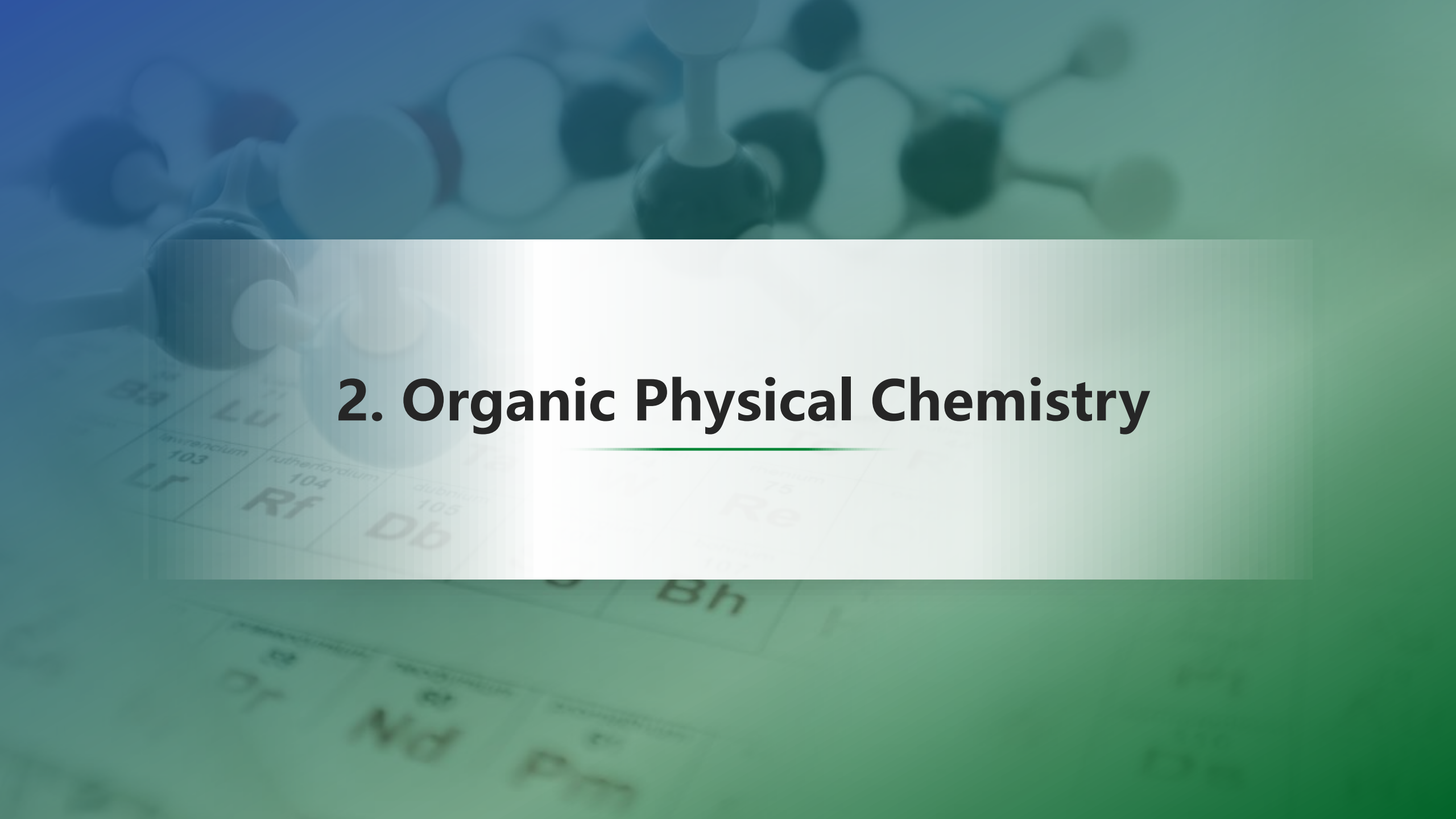
构效关系：性质匹配
(structure-affinity relationship, SAR)

Medicinal chemistry origin of QSAR

- In 1868, chemist Alexander Crum Brown and physiologist Thomas R. Fraser studied the biological effects of methylation of the alkaline N atoms of alkaloids.
- They proposed that the physiological activity of compounds depends on the composition of their components, i.e., the biological activity ϕ is a function of the composition C of the compound: $\phi=f(C)$, which is the famous **Crum-Brown equation**. This hypothesis laid the foundation for subsequent QSAR research.



Application-driven



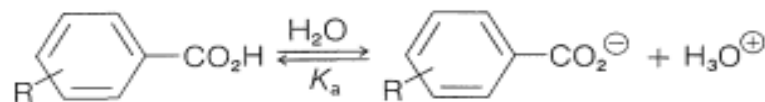
The background features a blurred molecular model with white and grey spheres connected by rods, overlaid on a portion of the periodic table. Visible elements include Lu, Lr, Rf, Db, Re, Bh, Nd, and Pt.

2. Organic Physical Chemistry

QSAR origin from organic physical chemistry

- the quantitative correlation of reaction rates and equilibria of organic reactions with the structure of the reactants
- compare the acid strengths (K_a) of a series of substituted benzoic acids with the strength of benzoic acid itself

Dissociation Constants ($10^{-5} \times K_a$) of Some Substituted Benzoic Acids in Water at 25°



R	H	CH ₃	OCH ₃	F	Cl	NO ₂
ortho	6.27	12.3	8.06	54.1	114	671
meta	6.27	5.35	8.17	13.6	14.8	32.1
para	6.27	4.24	3.38	7.22	10.5	37.0

QSAR origin from organic physical chemistry

- different reaction kinetics / thermodynamics can be related in terms of the effect of substitution groups

Specific Rate Constants^a for Alkaline Hydrolysis of Some Substituted Ethyl Benzoates in 85% Ethanol-Water Solution at 30°

$$\text{R-C}_6\text{H}_4\text{-CO}_2\text{C}_2\text{H}_5 + \text{OH}^\ominus \xrightarrow{k} \text{R-C}_6\text{H}_4\text{-CO}_2^\ominus + \text{C}_2\text{H}_5\text{OH}$$

R	H	CH ₃	OCH ₃	F	Cl	NO ₂
ortho	81.7	15.8		462	267	912
meta	81.7	57.7			605	5180
para	81.7	38.2	17.5	251	353	8480

^a10⁵k, liter mole⁻¹ sec⁻¹.

The linear correlation for meta and para substituents is observed for the rates or equilibrium constants for many reactions.

The straight line can be expressed by
 $\log k = \rho \log K + C$

For the ring substituent hydrogen (R=H),
 the Equation becomes
 $\log k_0 = \rho \log K_0 + C$

subtracting the two equations
 $\log(k/k_0) = \rho \log(K/K_0)$
 or $\log(k/k_0) = \rho \sigma$
 in which σ is defined as:
 $\sigma = \log(K/K_0)$

Hammett equation

Linear free energy relationship

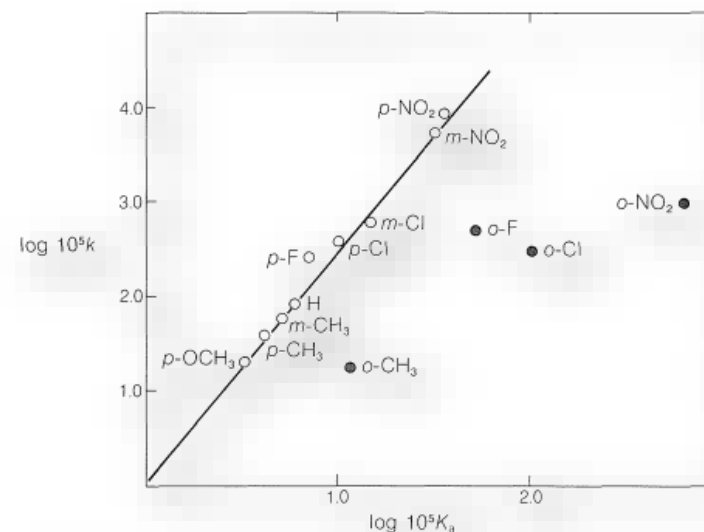


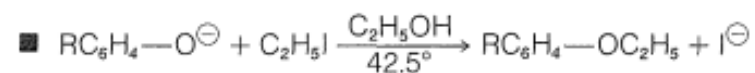
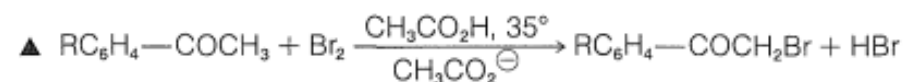
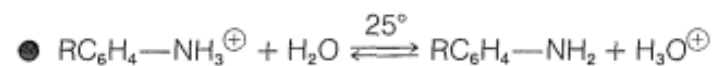
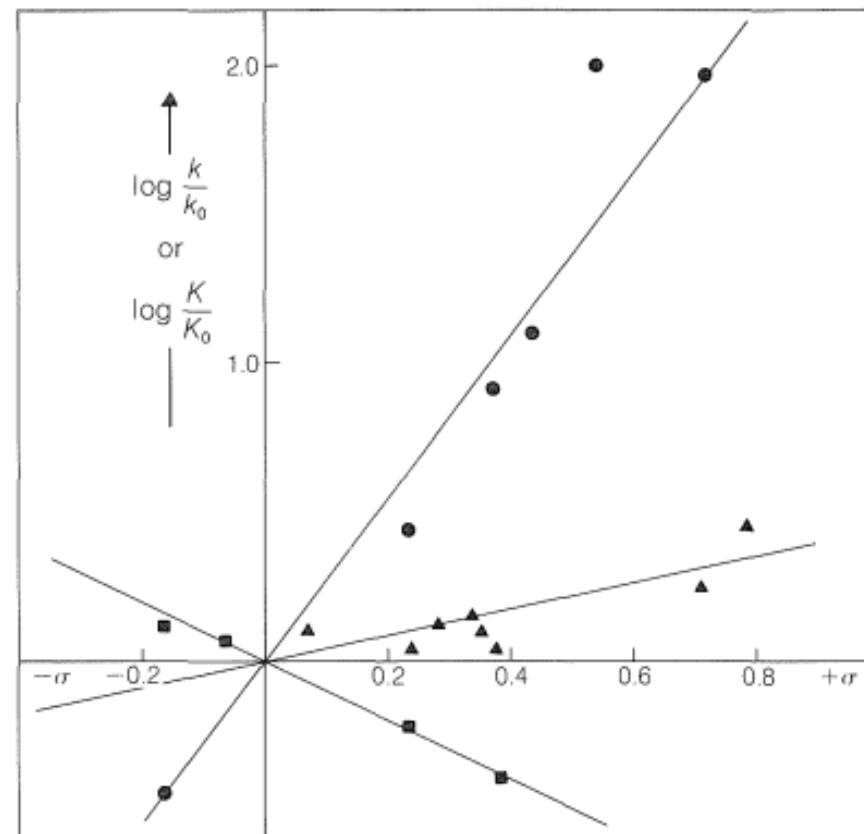
Figure 26-3: Plot of $\log 10^5 K_a$ for the dissociation of substituted benzoic acids in water at 25° against $\log 10^5 k$ for the rates of alkaline hydrolysis of substituted ethyl benzoates in 85% ethanol-water at 30°.

Table 26-6: Hammett Substituent Constants

σ			σ		
Substituent	Meta	Para	Substituent	Meta	Para
$\text{O}^{\ominus}$	-0.71	-1.00	F	+0.34	+0.06
OH	+0.12	-0.37	Cl	+0.37	+0.23
OCH_3	+0.12	-0.27	CO_2H	+0.36	+0.41
NH_2	-0.16	-0.66	COCH_3	+0.38	+0.50
CH_3	-0.07	-0.17	CF_3	+0.43	+0.54
$(\text{CH}_3)_3\text{Si}$	-0.04	-0.07	NO_2	+0.71	+0.78
C_6H_5	+0.06	-0.01	$(\text{CH}_3)_3\text{N}^{\oplus}$	+0.88	+0.82
H	0.00	0.00	$\text{N}_2^{\oplus}$	+1.76	+1.91
SH	+0.25	+0.15	$(\text{CH}_3)_2\text{S}^{\oplus}$	+1.00	+0.90
SCH_3	+0.15	0.00			

Different reaction has different ρ values

- A large ρ constant, positive or negative, means a high sensitivity to substituent influences.
- Reactions that are assisted by **high electron density** at the reaction site characteristically have **negative ρ values**, whereas reactions that are favored by **withdrawal of electrons** from the reaction site have **positive ρ values**.



No.	Equilibria	ρ
1	$\text{RC}_6\text{H}_4\text{—CO}_2\text{H} \xrightleftharpoons[25^\circ]{\text{H}_2\text{O}} \text{RC}_6\text{H}_4\text{—CO}_2^\ominus + \text{H}^\oplus$	1.00
2	$\text{RC}_6\text{H}_4\text{—CH}_2\text{CO}_2\text{H} \xrightleftharpoons[25^\circ]{\text{H}_2\text{O}} \text{RC}_6\text{H}_4\text{—CH}_2\text{CO}_2^\ominus + \text{H}^\oplus$	0.49
3	$\text{RC}_6\text{H}_4\text{—NH}_3^\oplus \xrightleftharpoons[25^\circ]{\text{H}_2\text{O}} \text{RC}_6\text{H}_4\text{—NH}_2 + \text{H}^\oplus$	2.77
4	$\text{RC}_6\text{H}_4\text{—CHO} + \text{HCN} \xrightleftharpoons[95\%]{\text{C}_2\text{H}_5\text{OH}} \text{RC}_6\text{H}_4\text{—CH(OH)CN}$	−1.49
5	$\text{RC}_6\text{H}_4\text{—OH} \xrightleftharpoons[25^\circ]{\text{H}_2\text{O}} \text{RC}_6\text{H}_4\text{—O}^\ominus + \text{H}^\oplus$	2.11

No.	Reaction rates	ρ
6	$\text{RC}_6\text{H}_4\text{—CO}_2\text{C}_2\text{H}_5 + \text{OH}^\ominus \xrightarrow[30^\circ]{85\% \text{ C}_2\text{H}_5\text{OH}} \text{RC}_6\text{H}_4\text{—CO}_2^\ominus + \text{C}_2\text{H}_5\text{OH}$	2.43
7	$\text{RC}_6\text{H}_4\text{—CO}_2\text{H} + \text{CH}_3\text{OH} \xrightarrow[25^\circ]{\text{H}^\oplus} \text{RC}_6\text{H}_4\text{—CO}_2\text{CH}_3 + \text{H}_2\text{O}$	-0.23
8	$\text{RC}_6\text{H}_4\text{—OH} + \text{C}_6\text{H}_5\text{COCl} \xrightarrow{20^\circ} \text{RC}_6\text{H}_4\text{—OCOC}_6\text{H}_5 + \text{HCl}$	0.56
9	$\text{RC}_6\text{H}_4\text{—O}^\ominus + \text{C}_2\text{H}_5\text{I} \xrightarrow[42.5^\circ]{\text{C}_2\text{H}_5\text{OH}} \text{RC}_6\text{H}_4\text{—OC}_2\text{H}_5 + \text{I}^\ominus$	-0.99
10	$\text{RC}_6\text{H}_4\text{—CH}_2\text{Cl} + \text{OH}^\ominus \xrightarrow[30^\circ]{\text{H}_2\text{O}} \text{RC}_6\text{H}_4\text{—CH}_2\text{OH} + \text{Cl}^\ominus$	-0.33
11	$\text{RC}_6\text{H}_4\text{—CH}_2\text{Cl} + \text{H}_2\text{O} \xrightarrow[30^\circ]{48\% \text{ C}_2\text{H}_5\text{OH}} \text{RC}_6\text{H}_4\text{—CH}_2\text{OH} + \text{HCl}$	-2.18
12	$\text{R—} \langle \text{benzene ring} \rangle \text{—CH(Cl)—C}_6\text{H}_5 + \text{C}_2\text{H}_5\text{OH} \xrightarrow{25^\circ} \text{R—} \langle \text{benzene ring} \rangle \text{—CH(OC}_2\text{H}_5\text{)—C}_6\text{H}_5 + \text{HCl}$	-5.09
13	$\text{RC}_6\text{H}_4\text{—NH}_2 + \text{C}_6\text{H}_5\text{COCl} \xrightarrow[26^\circ]{\text{C}_6\text{H}_6} \text{RC}_6\text{H}_4\text{—NHCOC}_6\text{H}_5 + \text{HCl}$	-2.78
14	$\text{RC}_6\text{H}_4\text{—COCH}_3 + \text{Br}_2 \xrightarrow[\text{CH}_3\text{CO}_2^\ominus]{\text{CH}_3\text{CO}_2\text{H}, 35^\circ} \text{RC}_6\text{H}_4\text{—COCH}_2\text{Br} + \text{HBr}$	0.42
15	$\text{RC}_6\text{H}_4\text{—H} + \text{NO}_2^\oplus \xrightarrow[18^\circ]{(\text{CH}_3\text{CO})_2\text{O}} \text{RC}_6\text{H}_4\text{—NO}_2 + \text{H}^\oplus$	-5.93
16	$\text{R—} \langle \text{benzene ring} \rangle \text{—Br} + \langle \text{piperidine ring} \rangle \xrightarrow{25^\circ} \text{R—} \langle \text{benzene ring} \rangle \text{—N} \langle \text{piperidine ring} \rangle + \text{HBr}$	4.92

Steric effect

Taft parameter E_s

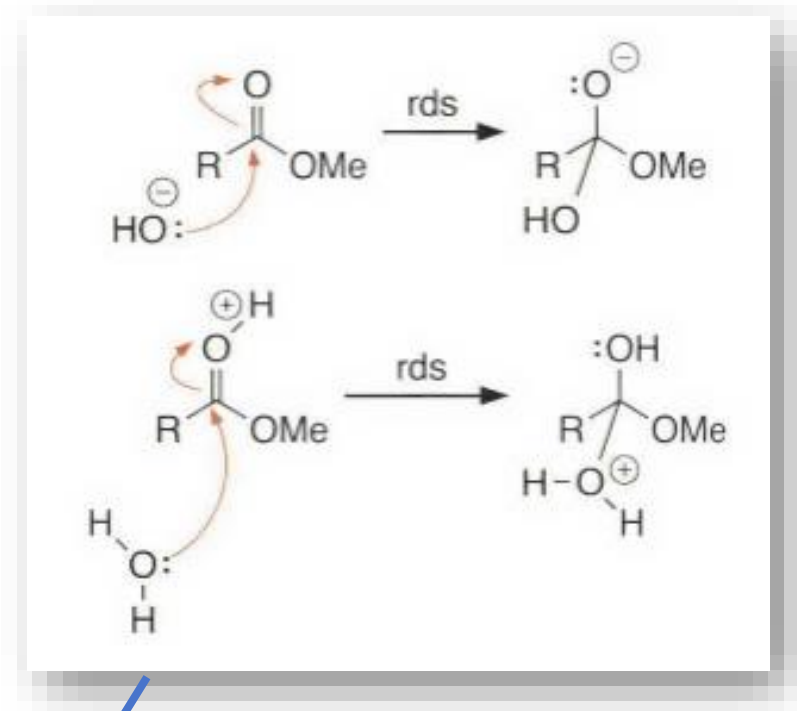
$$\log \left(\frac{k_s}{k_{\text{CH}_3}} \right) = \rho^* \sigma^* + \delta E_s$$

Steric effect

the ratio of the rate of the substituted reaction compared to the reference

Polar effect

$$\sigma^* = \left(\frac{1}{2.48\rho^*} \right) \left[\log \left(\frac{k_s}{k_{\text{CH}_3}} \right)_B - \log \left(\frac{k_s}{k_{\text{CH}_3}} \right)_A \right]$$

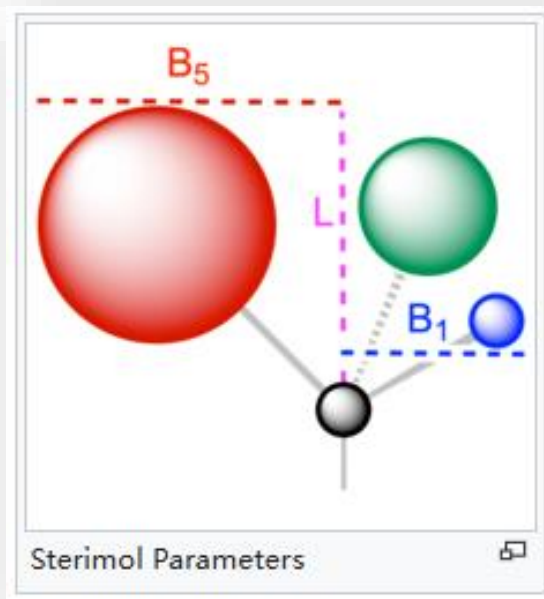


Difference at basic and acidic condition, the same steric effect

Steric effect

Taft parameter $E_s \rightarrow$ Charton (van der Waals radii)

Sterimol parameters containing two width parameters (B_1 and B_5) and a length parameter (L) and calculated by a program



Atomic attributed molecular properties

- **Molar refractivity (MR)**, related to polarizability
- A measure of the volume occupied by an atom or group of atoms

$$\text{MR} = \frac{(n^2 - 1)}{(n^2 + 2)} \times \frac{\text{MW}}{d}$$

- d density, n refractive index

A series of **atomic MR** values were defined considering the local environment of the atom in the molecule. The **sum** of atomic MR equals the MR of the whole molecule.

Partition Coefficient related to polarity

logP: hydrophobic of compounds

π : hydrophobic of substituent

	CH ₃	OH	Cl	F
π (aliphatic substituent)	0.50	-1.16	0.39	-0.17
π (aromatic substituent)	0.52	-0.67	0.71	0.14

$$\pi = \log P_x - \log P_H$$

P_H = parent compound

P_x = substituted compound.

知乎 @梳延斋主人

Partition Coefficient

$$\log P = \log \left(\frac{C_o}{C_w} \right)$$

High P value : **hydrophobic**

Low P value : **hydrophilic**



log P can also be attributed to atoms

Hansch model

- In 1964, Hansch and Fujita proposed the famous Hansch model, which pointed out that the biological activity of molecules is primarily determined by their hydrophobic effect ($\log P$), stereoelectronic effect (E_s), and electrostatic effect (σ), and assumed that these three effects can be independently added together, with the complete form being:
- $\log(1/C) = \alpha \log P + \beta E_s + \theta \sigma + \eta$

Atomic charge and EState

- NBO charge (NBO analysis after DFT calculation)
- Gasteiger Charge (rdkit)
- Estate_VSA VSA_Estate: MOE-type descriptors using EState indices and surface area contributions

Global molecular properties

- pKa
- Redox potential
- HOMO LUMO
- TPSA: topological surface area
- MolWt: molecular weight
- NumHAcceptors: number of hydrogen acceptors
- NumHDonors: number of hydrogen donors
- NumRotatableBonds: number of rotatable bonds
- Various rdkit descriptors: [Getting Started with the RDKit in Python — The RDKit 2023.03.1 documentation](#)

Vibrational frequency of a bond

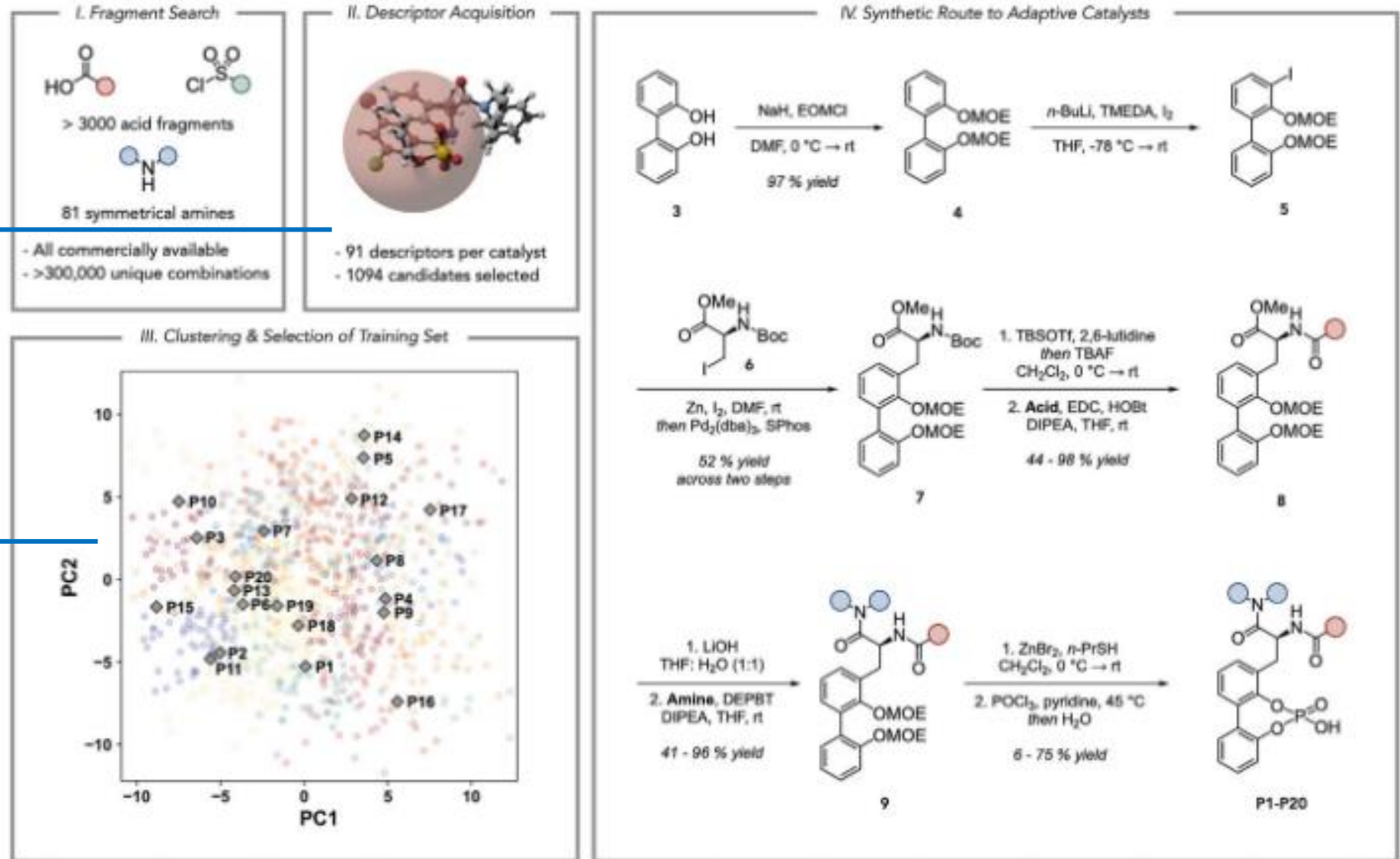
- Vibrational frequency of C=O bond reflects electronic structure
- Similarly, NMR chemical shift (usually ^{13}C NMR) reflects electronic structure

Example: Data science enables the development of a new class of chiral phosphoric acid catalysts 2023 Chem

Matthew Sigman

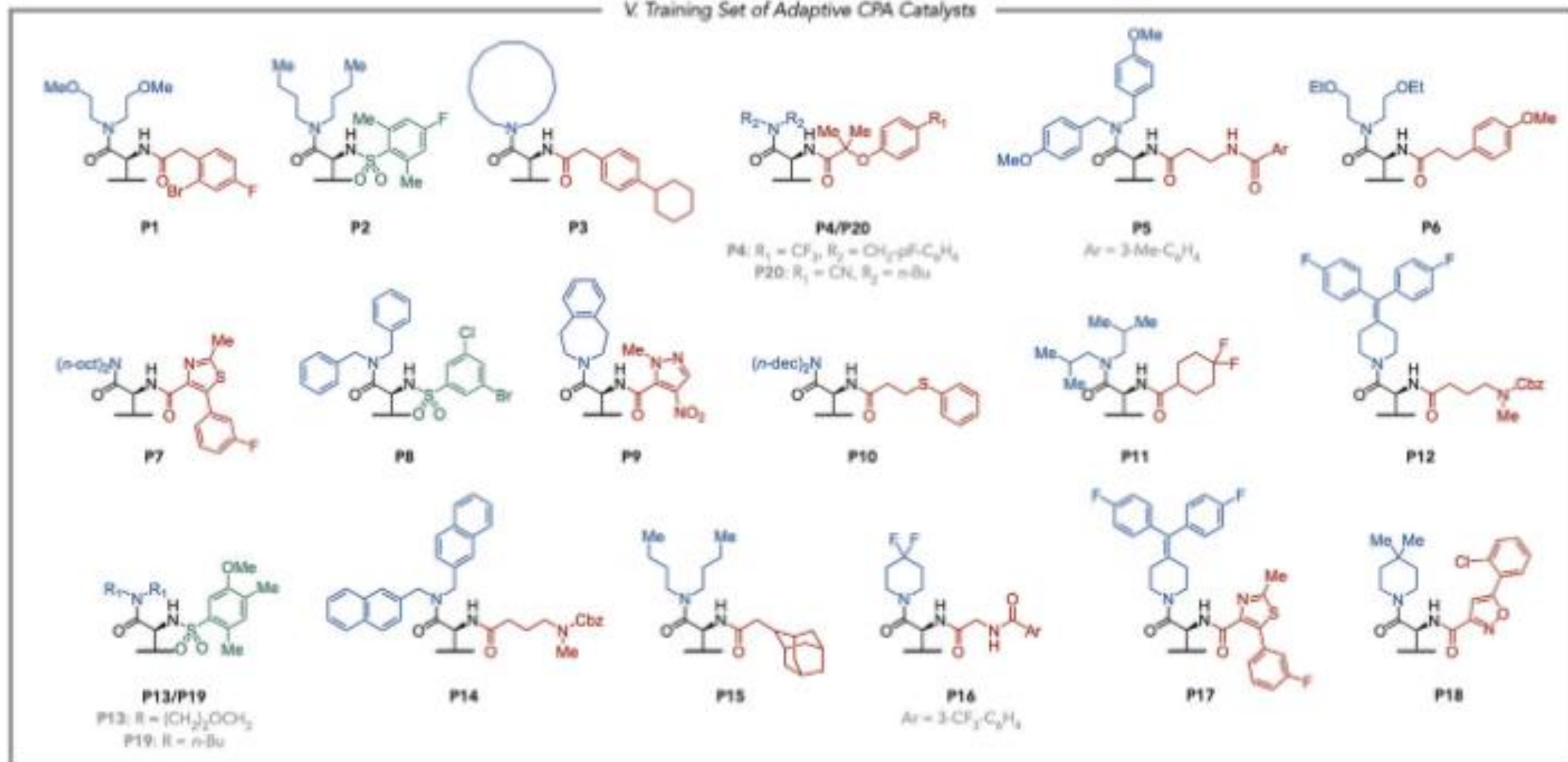
Descriptors for
substitutions of
the catalysts

Dimension
reduction for
visualization
(PCA, t-SNE,
UMAP)



Diverse structures

V. Training Set of Adaptive CPA Catalysts

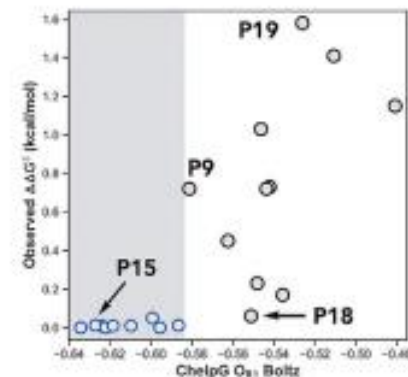
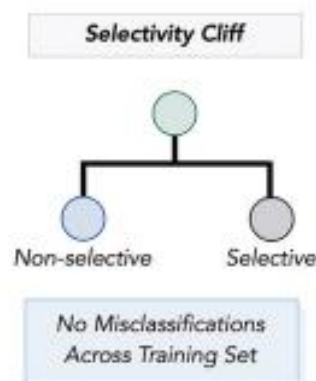


A single descriptor to correlate with ee values: partial charge on O

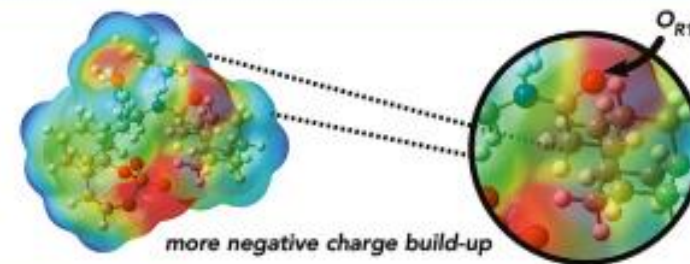
Parameter selection:

- Correlation
- Rule extraction (RIT)
- Stepwise regression

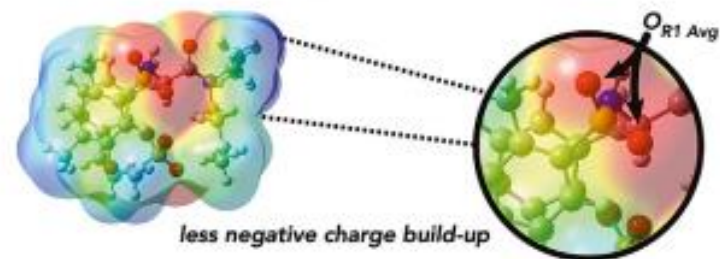
A Univariate Selectivity Classification



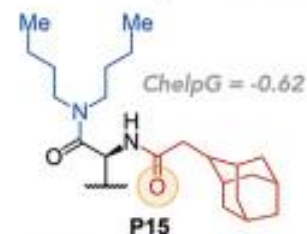
non-selective catalyst



highly-selective catalyst



B Partial Charge Visualization



All sulfonamides $\geq 85:15$ er

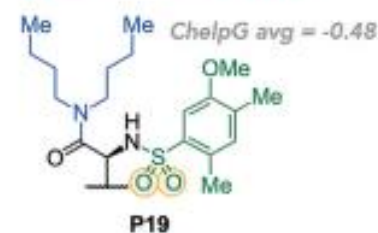


Figure 3. Univariate classification of selective vs. non-selective catalysts in training set

(A) Classification of catalysts reveals single partial charge parameter (ChelpG O_{R1}) as a predictor for observed activity cliff (threshold set at 52.5:47.5 er, 5% ee).

(B) Visualization of the partial charge using ESP maps. Non-selective and selective examples are shown.

Regression model

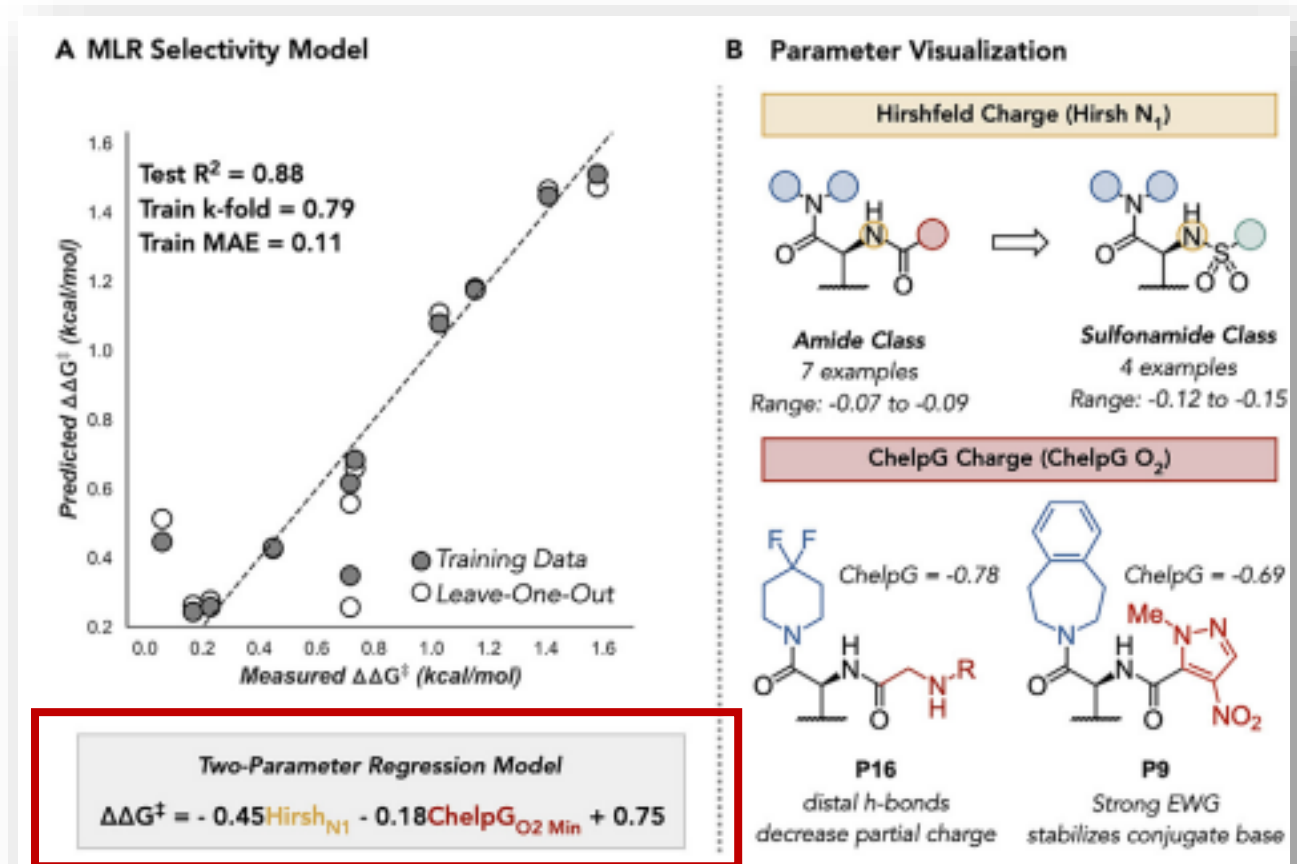


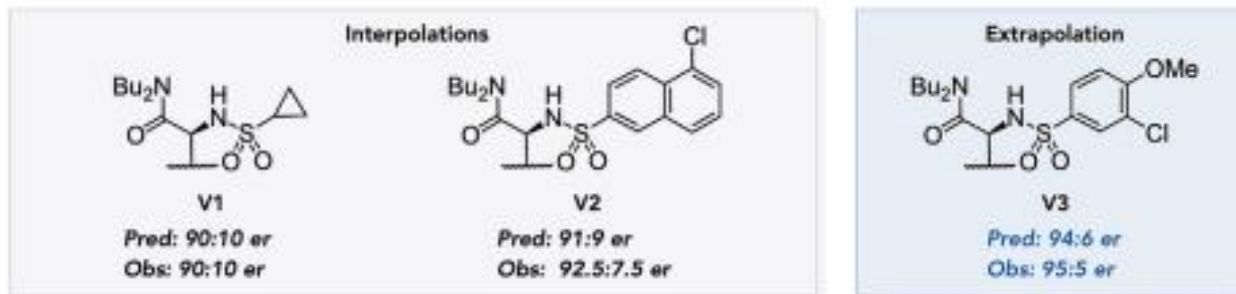
Figure 4. Multivariate linear regression model identifies features key for selectivity

(A) The statistical model reveals two key catalyst parameters that influence selectivity: Hirshfeld charge on N1 (secondary amide) and ChelpG on the O2 position of phosphate.

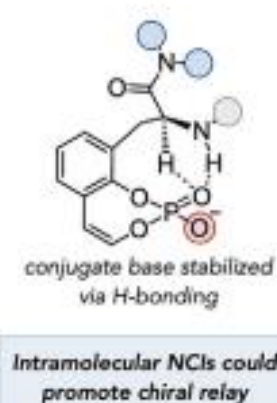
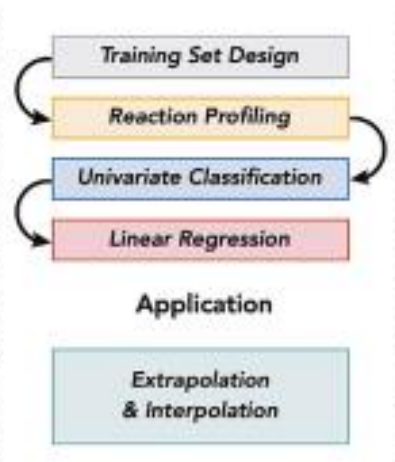
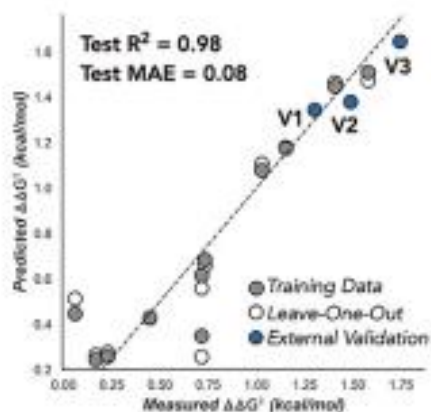
(B) Key catalyst features for selectivity from the regression model. Two partial charges give evidence changes in observed enantioselectivity related to intramolecular hydrogen-bonding networks.

Validate and apply the model

A External Validation Catalyst Predictions



B Model Validation & Design Workflow



Avoid data leakage

Train Validation Test

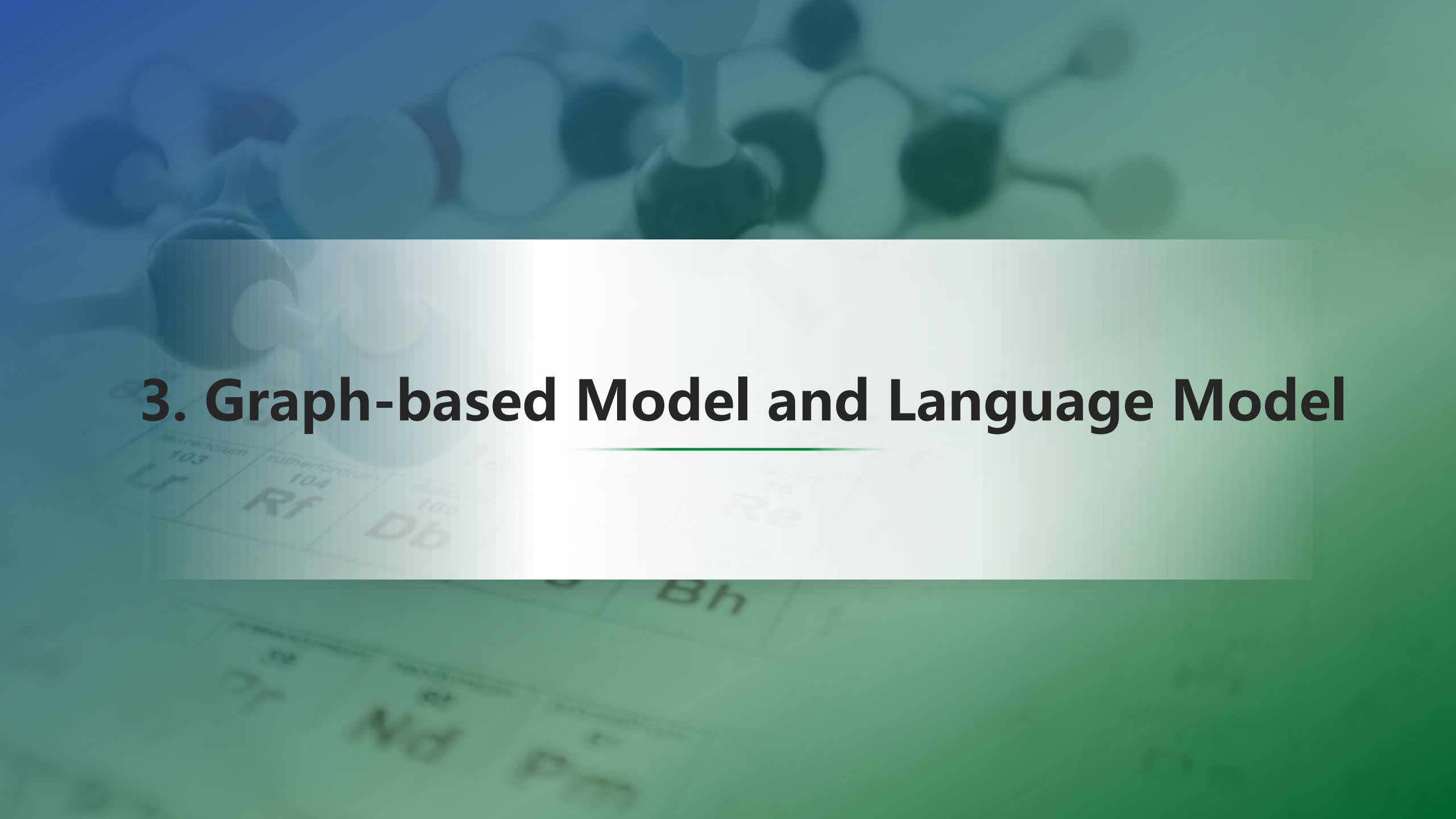
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Model fitting Hyper-parameter exam

Compare to baseline model

Note: Feature Selection is a special type of hyper-parameter selection

Figure 5. External validation of statistical model through interpolation and extrapolation
 (A) Expanded test set of sulfonamide catalysts reveals opportunity for extrapolation (V3). Two additional interpolations were also predicted to validate the established MLR model.
 (B) Validated model with out-of-set predictions plotted. Workflow for analyzing adaptable CPA scaffold uses data science from the outset to allow for accurate extrapolation and interpolation.



3. Graph-based Model and Language Model

2D-QSAR

- SMILES
- Fingerprints
- Molecular graph

SMILES

- SMILES (Simplified Molecular Input Line Entry System) is a notation system used to represent chemical structures in a linear string format. It provides a simple and compact way to describe the connectivity and arrangement of atoms in a molecule.
- the SMILES string for water is "O",
- for ethanol, it's "CCO".

- Explicitly expressing functional group
- Directly fit into language model

Is GPT-3 all you need for machine learning for chemistry?

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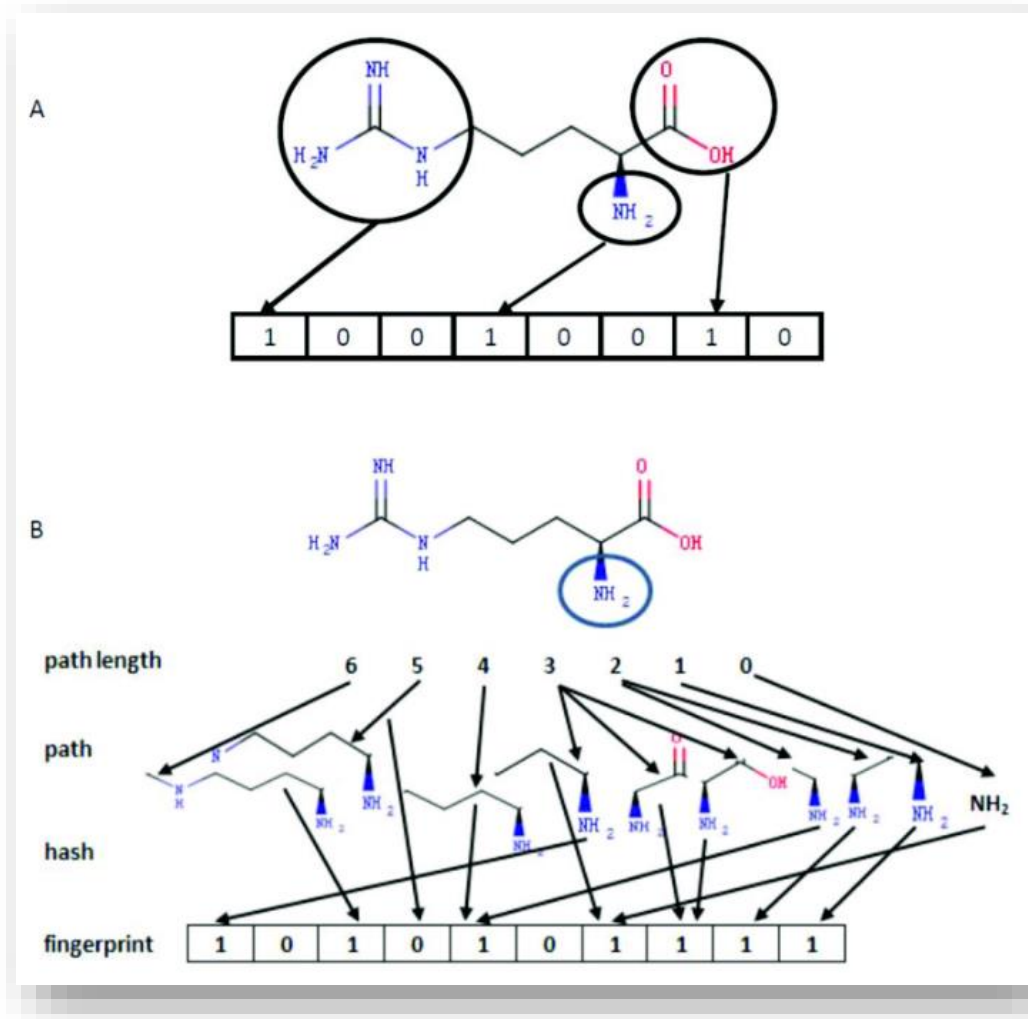
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Table 1: Classification metrics for the photoswitch case study. Values indicate the means and standard deviation of at least three (typically ten) independent runs. The baseline is based on the GPR model proposed by Griffiths et al.¹⁹ The metrics for the MOF and dispersant case studies can be found in the Appendix in Table 3. For the TabPFN³⁰ baseline we used Morgan fingerprints with a bit size of only 100 as the current TabPFN model only has been pretrained for up to 100 features. For the MolCLR baseline fine-tuned a graph-isomorphism network (GIN, ~220 thousand parameters) pretrained by Xu et al.³¹ on ~10 million unique molecules.

model	accuracy (macro)	F ₁ micro	F ₁ macro
<i>photoswitch few shot (10 training points)</i>			
GPT-3 on IUPAC names	0.82 ± 0.03	0.64 ± 0.06	0.34 ± 0.08
GPT-3 on SMILES	0.75 ± 0.03	0.34 ± 0.12	0.17 ± 0.07
GPT-3 on SELFIES	0.76 ± 0.04	0.33 ± 0.11	0.18 ± 0.07
GPR ¹⁹	0.75 ± 0.04	0.40 ± 0.10	0.26 ± 0.15
TabPFN ³⁰	0.79 ± 0.03	0.44 ± 0.16	0.26 ± 0.10
MolCLR ³² , GIN ³¹	0.71 ± 0.04	0.28 ± 0.10	0.09 ± 0.03
<i>photoswitch (300 training points)</i>			
GPT-3 on SMILES	0.87 ± 0.05	0.66 ± 0.16	0.61 ± 0.12
GPT-3 on SELFIES	0.88 ± 0.01	0.70 ± 0.03	0.62 ± 0.11
GPR ¹⁹	0.91 ± 0.02	0.78 ± 0.05	0.77 ± 0.07
TabPFN ³⁰	0.88 ± 0.02	0.71 ± 0.05	0.66 ± 0.10
MolCLR ³² , GIN ³¹	0.90 ± 0.01	0.75 ± 0.03	0.69 ± 0.08

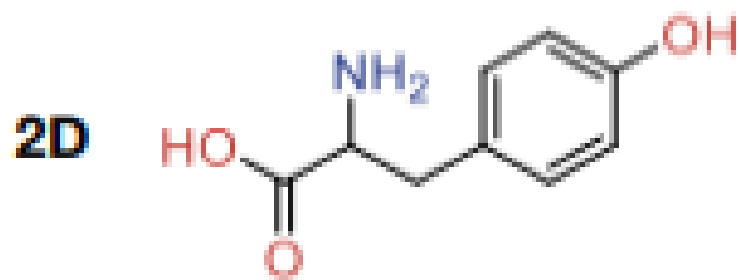
Fingerprints

- Analyzing molecules to their constitutional parts
- Encoding the structural information as a string of binary digits (bits) or as a vector of numbers, capturing various aspects like the presence of particular substructures, functional groups, or even more abstract properties like molecular shape.
- MACCSKeys, Morgan Fingerprint
- Moffi, MFF



Graph

Mathematical graph: atoms to be the node and bonds to be edges

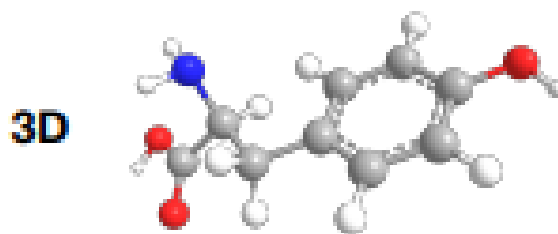


Message passing neural network
GCN

Table 1: Details of node and edge features in molecular graphs.

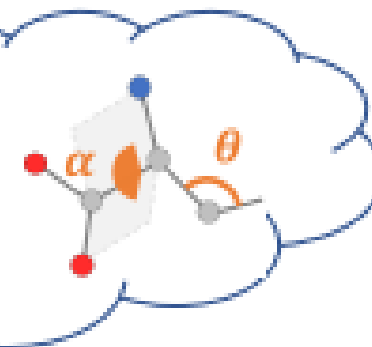
Attribute	Details
Node	
Atom type	118
Chirality tag	unspecified, tetrahedral cw, tetrahedral ccw, other
Hybridization	sp, sp ² , sp ³ , sp ³ d, or sp ³ d ²
Atomaricity	0 or 1 (aromatic atom or not)
Edge	
Bond type	single, double, triple, aromatic
Ring	0 or 1 (bond is in a ring or not)
Bond direction	-, endupright, enddownright
Stereochemistry	-, any, Z, E, cis, trans, @

Graph for 3D
representation



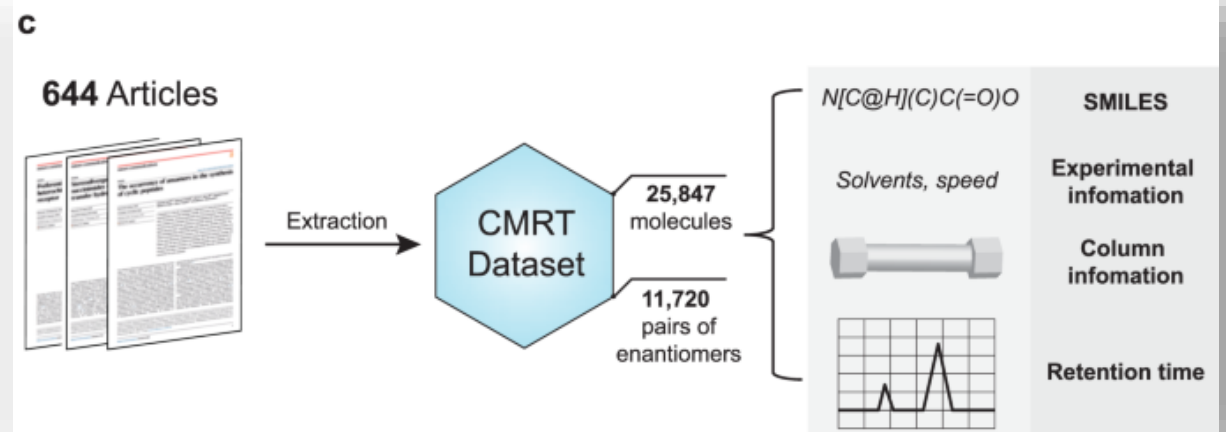
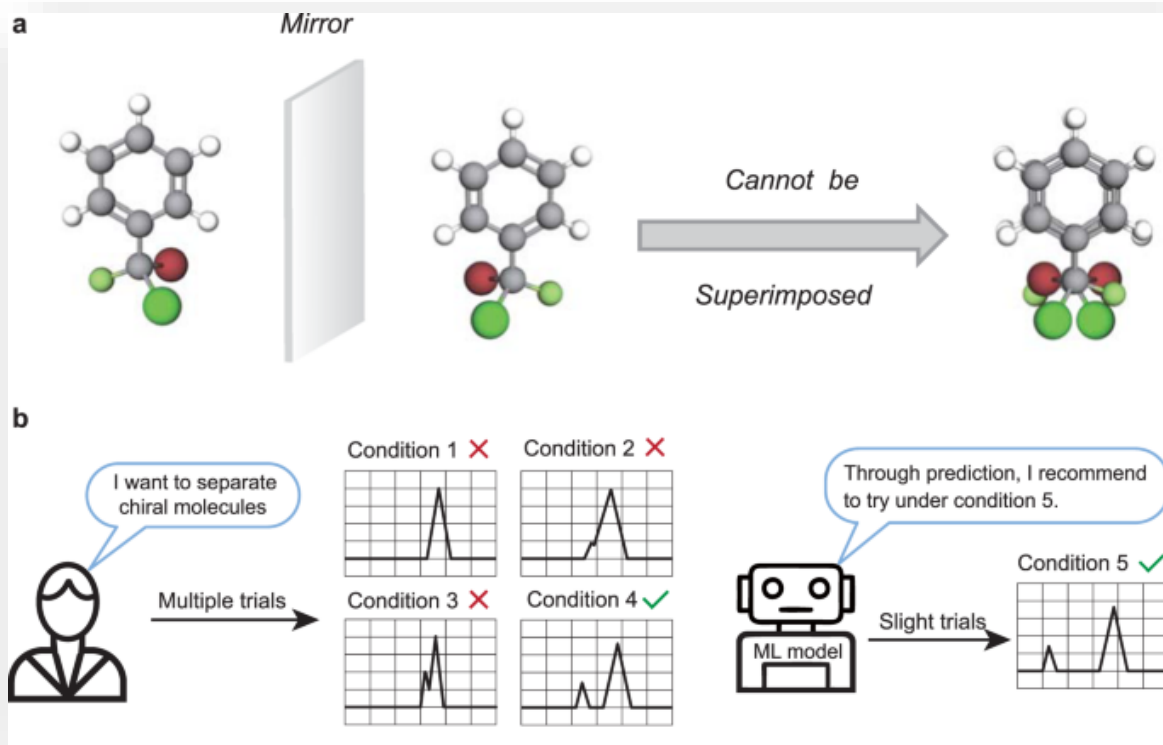
Directed-GNN
Geometric MPNN
(Node: atom)

Dihedral angle
Bond angle



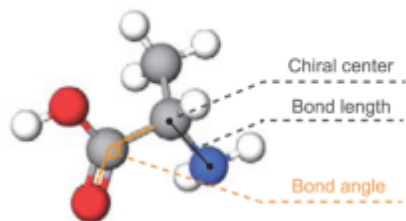
Example: Retention time prediction for chromatographic enantioseparation by quantile geometry-enhanced graph neural network

Fanyang Mo, Peking University



a

3D Conformation



Graph G

Node: Atom Edge: Bond

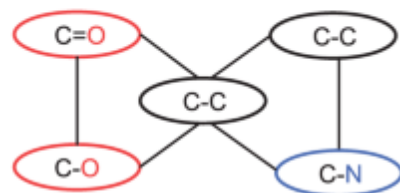


Atom feature

Atomic number	Explicit valence	
Formal charge	Implicit valence	
Hybridization	Aromaticity	Degree
Connected hydrogens	Chiral tag	

Graph H

Node: Bond length Edge: Bond angle



	O	C	O	C	C	N
O						
C						
O						
C						
C						
N						

Bond feature

Bond direction	Bond type
Whether in the ring or not	

Graph H feature

Bond length	Bond angle
-------------	------------

One graph from atomic node and bond edges

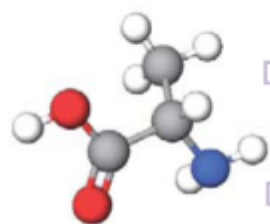
One graph from bond as node and bond angle as edges

b

Experimental settings

Proportion

MATS: moran coefficient

Graph G
(atom-bond)Graph H
(bond-angle)Graph
representation y^{90}
(90th quantile) y^{pred}
($R1 \times Speed$) y^{10}
(10th quantile)Molecular
descriptors

TPSA

RPSA

RASA

MEDC

MATS

TPSA : total polar surface area

RPSA: relative polar surface area

RASA: relative hydrophobic surface area

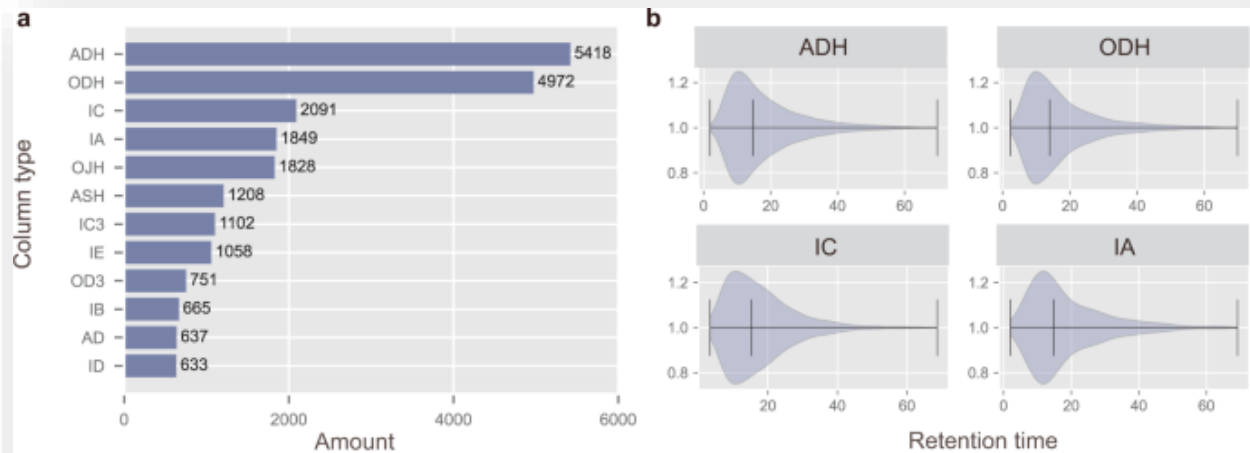
MEDC: molecular distance edge

Experimental setting as a vector

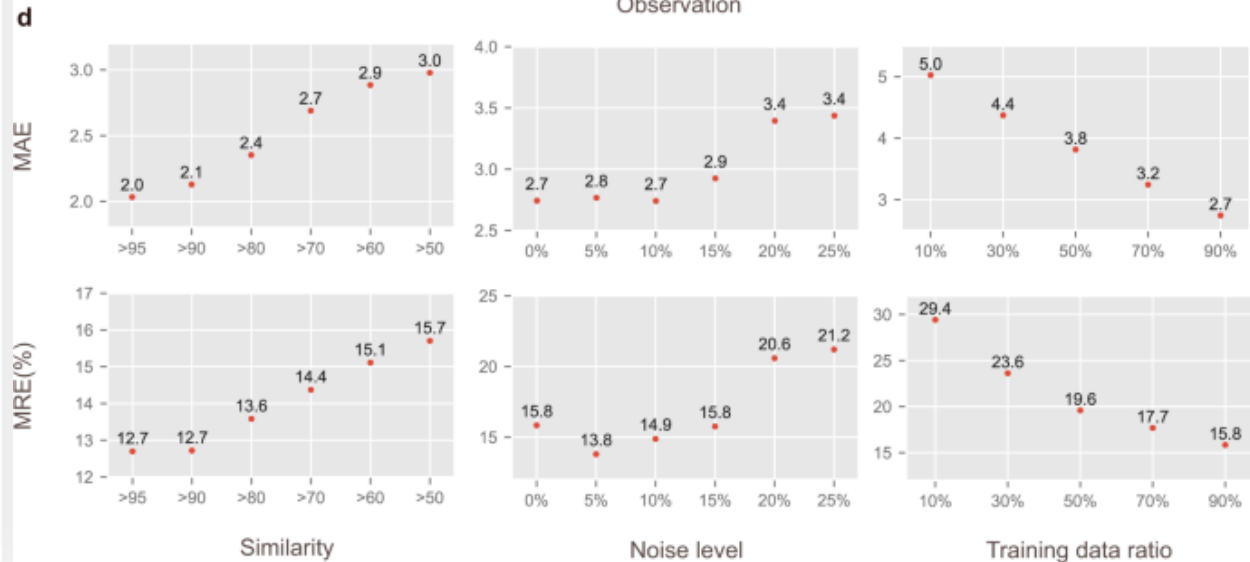
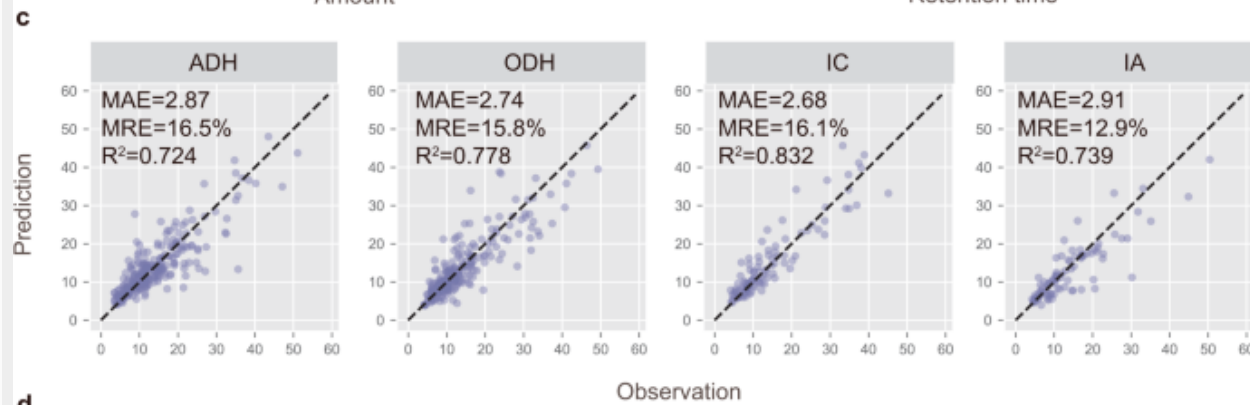
GCN will give a vector

Molecular descriptor as a vector

combine

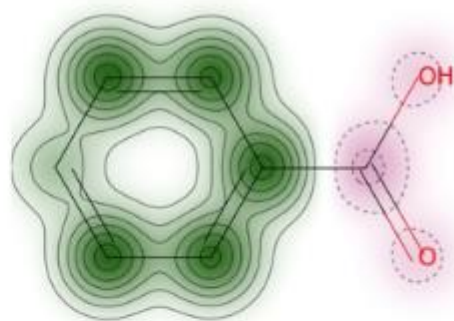


Information of different columns



SPOC

(an ensemble structure and physicochemical descriptor)



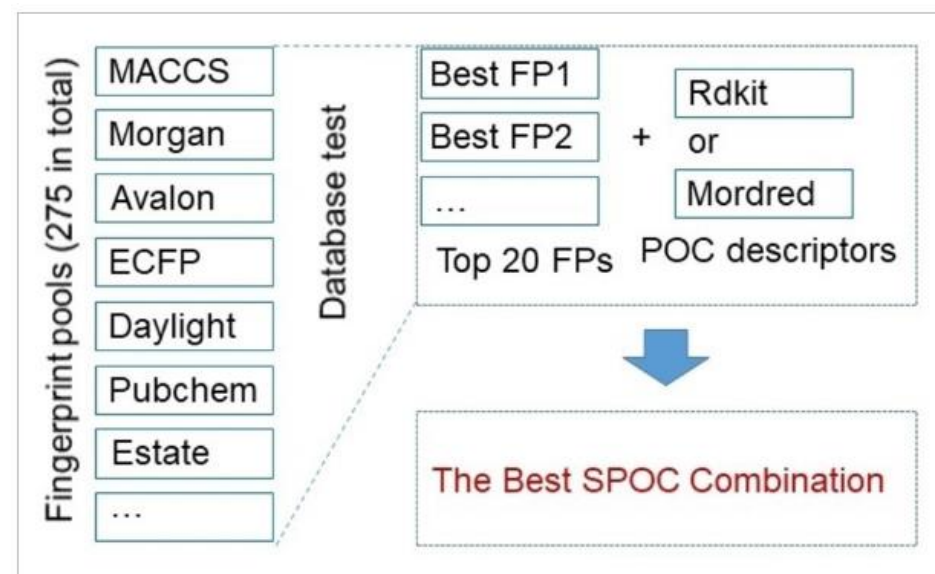
Structure:
Molecular Fingerprints

+

Physical organic chemistry
descriptor

Universal Ensemble Descriptor for AIChem

Luo Sanzhong, Tsinghua



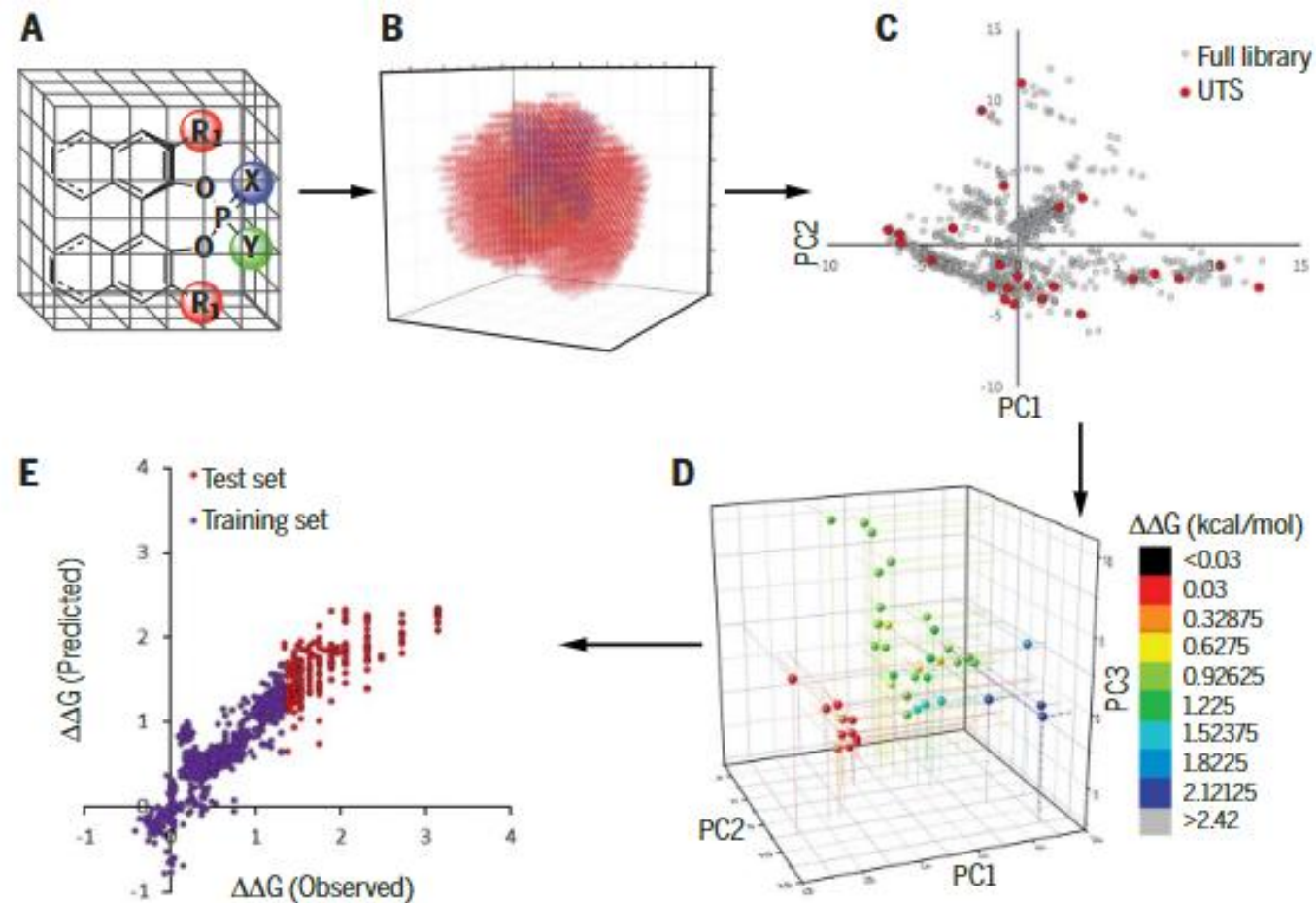
3D-QSAR

- The Comparative Molecular Field Analysis (CoMFA) is a widely used 3D-QSAR model.
- CoMFA calculates the forces acting on various positions in the space where the molecule exists (usually selected by grid methods) to characterize the three-dimensional structure of the molecule.
- There are also some other attempts in the field, including characterization methods through electron density, three-dimensional molecular images, or adding geometric information to molecular graphs.

Example: Prediction of higher-selectivity catalysts by computer-driven workflow and machine learning

2019, Science

Scott E. Denmark



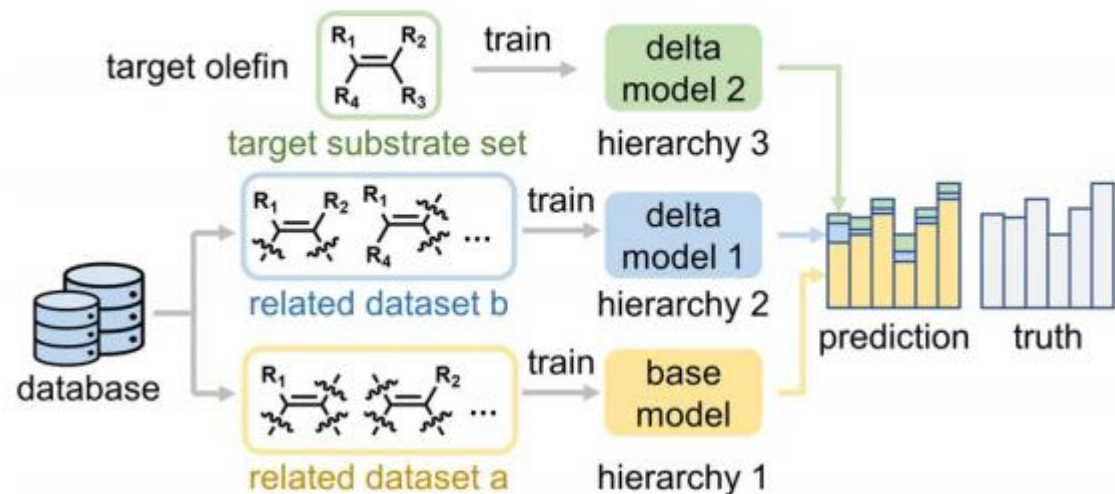
Cheminformatics-guided optimization protocol. (A) Generation of a large in silico library of catalyst candidates. (B) Calculation of robust chemical descriptors. (C) Selection of a UTS. (D) Acquisition of experimental selectivity data. (E) Application of machine learning to use moderate- to low-selectivity reactions to predict high-selectivity reactions. R, any group; X, O or S; Y, OH, SH, or NHTf; PC, principal component; $\Delta\Delta G$, mean selectivity.

3.5D – 4D

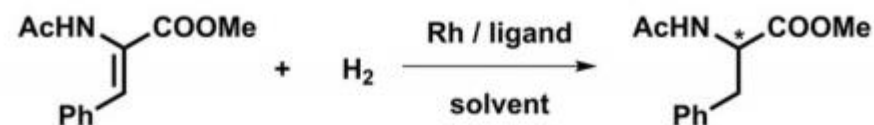
- One drawback of 3D representation is that a 3D structure only represents one **conformation/configuration** of the molecule. One molecule has multiple configurations and is vibrating in 3D space.
- Adding the time axis leads to the **4D** representation.
- Using multiple configurations for the representation constitutes the **3.5D** representation.

Example: Towards Data-Driven Design of Asymmetric Hydrogenation of Olefins: Database and Hierarchical Learning

a) Designed hierarchical learning approach

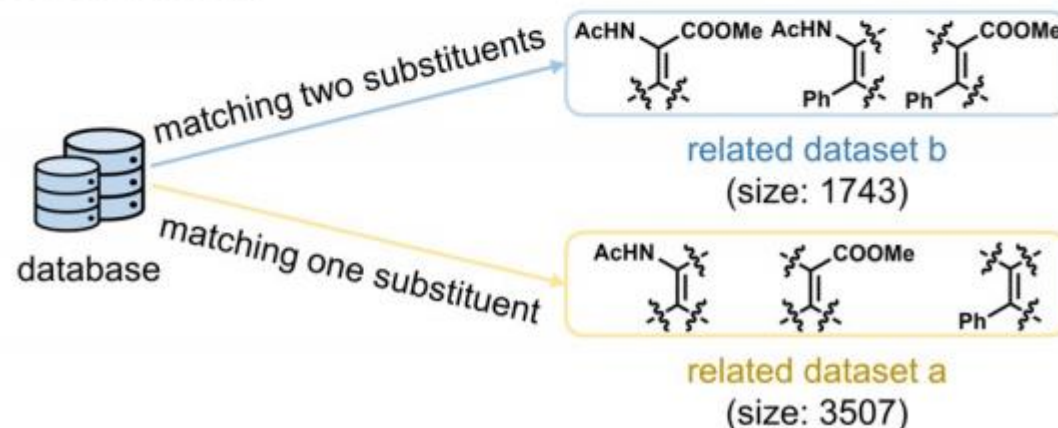


b) Illustration using methyl (Z)-2-acetamido-3-phenylacrylate

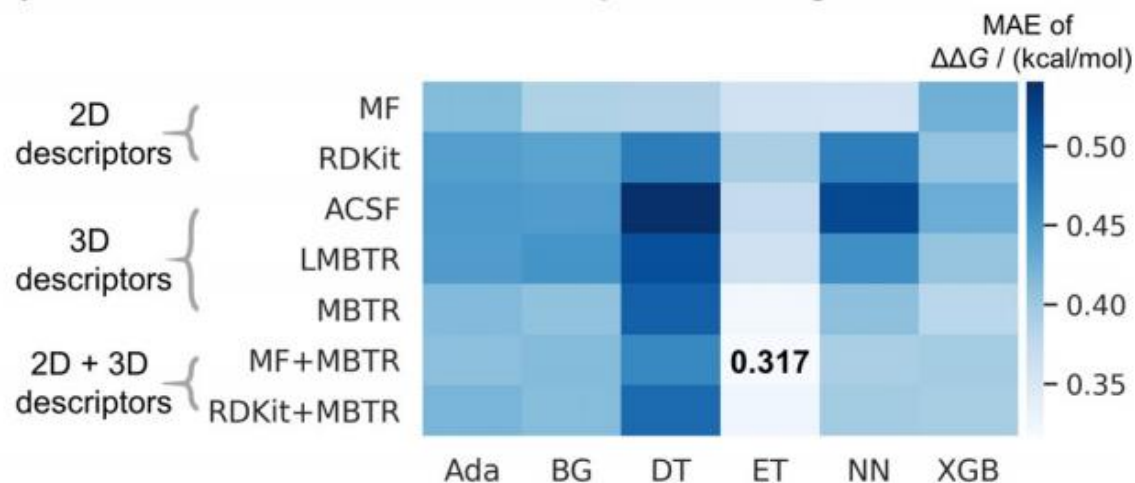


- 1 olefin
 - 59 ligands
 - 10 solvents
- 86 associated data
- 43 target substrate set
 - 43 test data

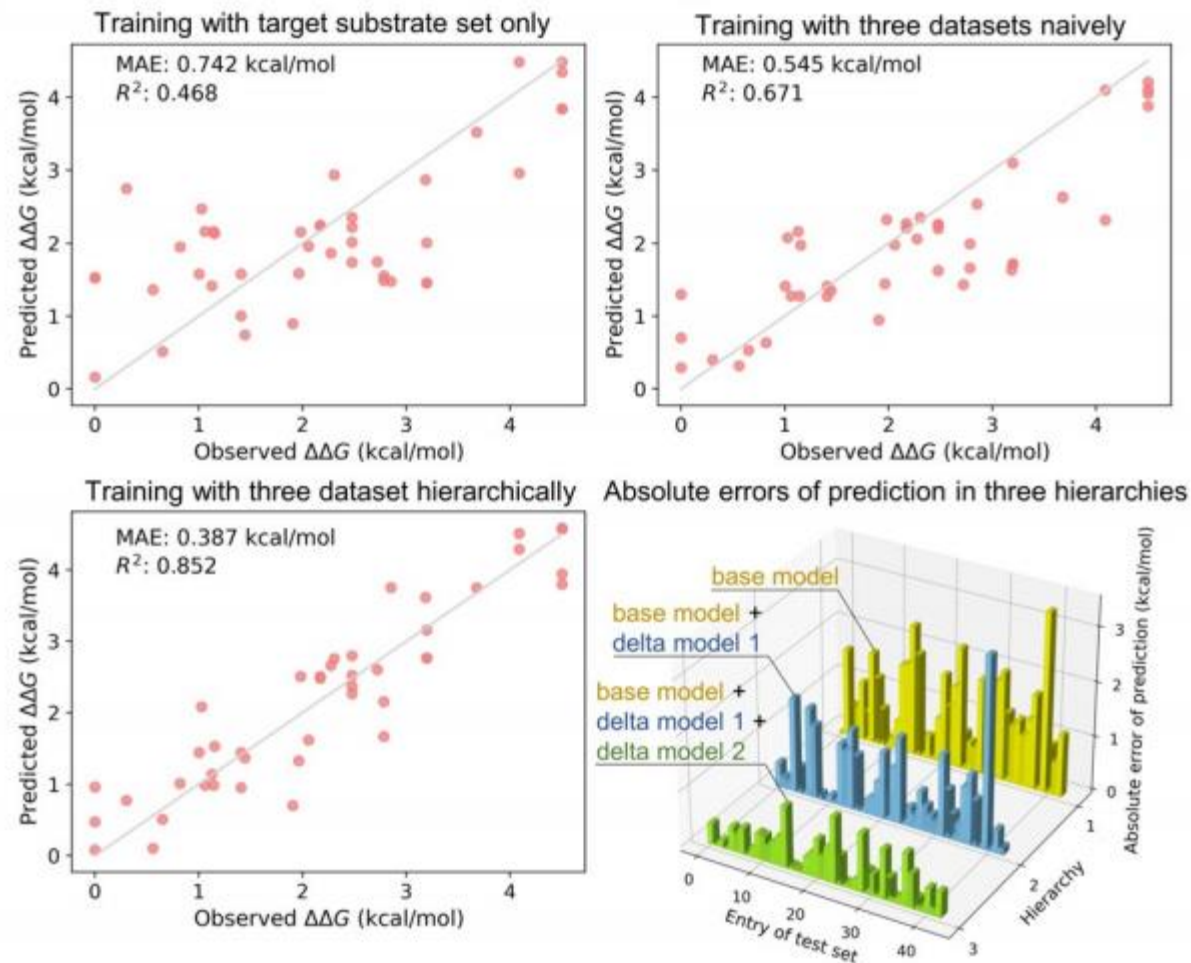
related dataset



c) Benchmark evaluation of descriptors and algorithms



d) Performance of hierarchical learning



Pre-trained model

- In the field of chemical research, one of the main challenges faced by QSAR modeling is the limited amount of data.
- Faced with this situation of insufficient labeled data, in more mature fields of machine learning such as Natural Language Processing (NLP) and Computer Vision (CV), the Pretrain-Finetune model has become a universal solution.
- Pretraining refers to initially training the model on a large amount of unlabeled data through self-supervised learning, enabling the model to acquire some basic information and general capabilities. Then, the model is fine-tuned on limited labeled data through supervised learning to equip it with problem-specific reasoning abilities.

Uni-Mol

What is Machine Learning ?

What is learning?

Herbert A. Simon →

Learning is any process by which a system improves performance from experience

Tom M. Mitchell →

Machine Learning is the study of algorithms that improve their performance P at some task T with experience E .



Herbert A. Simon

Nobel Prize

Turing Award

Decision Making



Tom M. Mitchell

American Academy of Arts and Sciences

National Academy of Engineering